

1. Overview

- a. I analyzed scene-level macro narrative features extracted from scripts using the provided macro pipeline outputs. Instead of only averaged values per movie, I focused on per-scene dynamics, distributions of narrative features, and clustering of movies based on tension.
- b. Goals:
 - i. Examining how narrative variables evolve across scenes
 - ii. Comparing distributions of variables across movies
 - iii. Identifying similarities in narrative pacing via clustering

2. Data Used

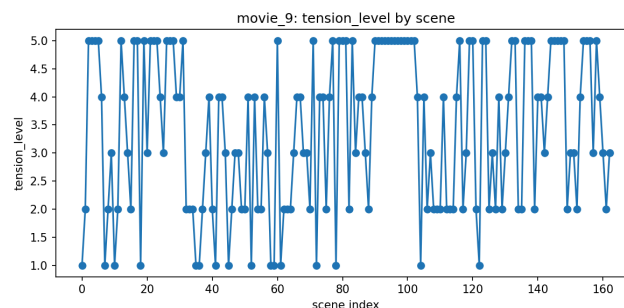
- a. For each movie, I used `scene_table.csv`, which contains per-scene values for:
 - i. `tension_level`
 - ii. `stakes_level`
 - iii. `irreversibility_level`
 - iv. `uncertainty_level`
 - v. `narrative_velocity`
 - vi. `dependency_on_prior_scenes`
 - vii. `resolution_progress`
- viii. **Note: I also used `all_movies_distribution_stats.csv` for cross movie comparisons and `tension_clusters.csv` for clustering results**

3. Method

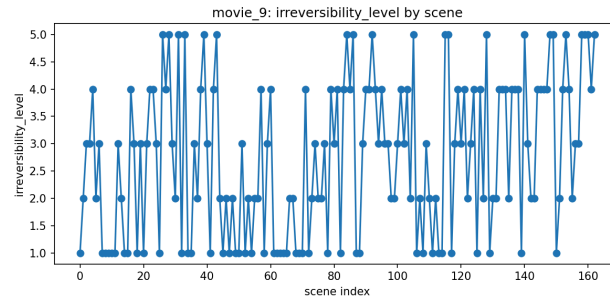
- a. For each movie:
 - i. plotted scene index vs:
 1. `tension_level`
 2. `irreversibility_level`
 3. `resolution_progress`
 - ii. computed distribution statistics:
 1. Mean, standard deviation, min, max
- b. For clustering:
 - i. Extracted `tension_level` sequences per movie
 - ii. Resampled all sequences to equal length
 - iii. Applied k-means clustering to group movies by tension patterns

4. Movie-Level Timelines

- a. Images



1.



2.

- ii. **Movie 9(12 and Holding Anthony Cipriano) shows a highly variable tension profile, with frequent jumps between low and high values rather than a smooth buildup. Irreversibility also fluctuates, but higher values appear more frequently in later scenes, suggesting increasing narrative consequence over time.**

5. Distribution Analysis

	A	B	C	D	E	F
1	movie	variable	mean	std	min	max
2	movie_1	tension_level	3.877551	1.271872	1	5
3	movie_1	stakes_level	4.040816	1.142128	1	5
4	movie_1	irreversibility_level	3.102041	1.249406	1	5
5	movie_1	uncertainty_level	3.591837	1.210808	1	5
6	movie_1	narrative_velocity	3.102041	1.249406	1	5
7	movie_1	dependency_on_prior_s	4.326531	0.934328	1	5
8	movie_1	resolution_progress	5.795918	13.88286	0	100
9	movie_10	tension_level	3.539568	1.450707	1	5
10	movie_10	stakes_level	3.690647	1.268596	1	5
11	movie_10	irreversibility_level	2.676259	1.368798	1	5
12	movie_10	uncertainty_level	3.194245	1.35114	1	5
13	movie_10	narrative_velocity	2.532374	1.219164	1	5
14	movie_10	dependency_on_prior_s	4.280576	0.839911	1	5
15	movie_10	resolution_progress	6.489209	18.28132	0	100

a.

- i. The sample shown here compares Movie 1 (Before We Go Ron Bass and Jen Smolka) and Movie 10(100 Feet Eric Red). For tension, Movie 1 has a higher mean (3.88) than Movie 10 (3.54), indicating a stronger average level of narrative pressure. However, Movie 10 has a higher standard deviation (1.45 vs 1.27), suggesting more fluctuation in tension across scenes.
- ii. A similar pattern appears for irreversibility. Movie 1 has a higher mean (3.10 vs 2.68), implying more consistently consequential scenes, while Movie 10 spans the full range but less consistently.
- iii. Movie 1 also shows higher average uncertainty (3.59 vs 3.19) and narrative velocity (3.10 vs 2.53), suggesting a more active and uncertain flow. Both movies have high dependency on prior scenes (above 4.2), indicating strong continuity.

- iv. Resolution progress varies more due to its 0–100 scale. Movie 10 has a slightly higher mean (6.49 vs 5.80) and greater spread (18.28 vs 13.88), indicating more uneven progression toward closure.
- v. Overall, the results show differences not just in average levels, but in variability, reinforcing that narrative structure should be analyzed as a distribution over time rather than a single aggregated value.

6. Tension Clustering

movie	cluster
movie_1	3
movie_10	1
movie_2	1
movie_3	0
movie_4	2
movie_5	1
movie_6	2
movie_7	1
movie_8	1
a. movie_9	0

- i. Clustering based on tension trajectories groups movies with similar narrative pacing patterns. some clusters show smoother progression, while others exhibit sharp fluctuations in tension.
- ii. This suggests that scripts can be grouped based on how tension evolves over time, not just overall intensity.

7. Key Findings

- a. Movies exhibit different tension structures, including gradual buildup and rapid oscillation
- b. Irreversibility does not always align directly with tension, indicating that not all high-tension scenes have lasting consequences
- c. Distribution statistics reveal differences in both intensity and variability across scripts
- d. Clustering identifies distinct narrative pacing styles across the dataset

8. Next Steps

- a. Compare vectors across decades using embedding distances
- b. Analyze relationships between macro features and genre or production context
- c. Incorporate additional vector components (summary, arc, numeric features)
- d. Explore deeper time-series clustering methods